# Philip Robinson

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**https://github.com/probinso**


10+ years experience in software, machine learning & DevOps on interdisciplinary teams supporting work across natural sciences, computer securities, remote sensing &|| citizen sciencery

**Skills/Experience**

| | | | |
|---|---|---|---|
| Machine Learning | Remote/Global Teams | Linux Systems | MLOps/DevOps |
| Data Sciences | Technology Transfer | Education/Mentoring | Rubber Ducking |

**Technologies**

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|---|---|---|---|
| Python/PyTorch/sklearn | Angular/Next.js/FastAPI | scipy.signal/librosa/sox | SQL/BQL/PostGIS |
| C/C++ | GDAL/cartopy/ESA-SNAP | AWS/GCP/firebase | Linux/Bash/Docker |

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**LLNL** — **Scientific Programmer (Nuclear)** — **Jan 2027 – present**

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**Blue Oasis** — **ML Engineer (Digital Twins)** — **Contract · May 2025 – Oct 2025**

- Improved ML edge multi-label acoustic event detection pipeline, for tracking vessels & marine life
- Full stack + security & performance on HydroTwin, a user focused acoustics analytics/IR dashboard
- Advised DL & surrogate model design on PHAROS & SmartFisher, for ecosystem monitoring & simulation

**Global Fishing Watch** — **Scientific Programmer (Earth Imaging)** — **May 2022 – March 2025**

- Implemented remote sensing pipeline for detecting vessels & matching to reported GPS locations
- Evaluated vessel detection over terrabytes of processed synthetic aperture radar (SAR) images
- Developed gfwsat to join & catalog millions of satellite scenes, serving global ocean surveys
- Developed simple offline satellite image labeler, supporting all shapely geometries by lat/lon
- Led technology transfer work, moving research prototypes to production & automated pipelines
- Developed collaboration strategies & best practices docs, for remote & international teams

**GrammaTech** — **Deep Learning Researcher II (Security)** — **March 2020 – April 2022**

- Research on binaries in vulnerability detection & improving code legibility from decompilation
- Developed feature extraction pipeline over local cluster & virtual machines on terrabytes of data
- Implemented, verified & incorporated ML/DL code from academic research into release products
- Participated in authoring & reviewing SBIR/STTR proposals for DOD/DARPA funding solicitations

**HappyWhale** — **Mobile & Full Stack Engineer** — **Contract · March 2019 – March 2020**

- Developed x-platform citizen science phone app for eco-tourism & wildlife population surveys
- Developed core tools for tracking marine life sightings & individual whale identification
- Contributed to data access API & web UI, supporting researchers in population ecology studies

**NASA's Jet Propulsion Lab** — **Data Science Intern (Natural Text)** — **June 2018 – Sept 2018**

- Prototyped employee expert recommender system, to eliminate weeks in ticket triage/assignment
- Designed & completed prototype from research papers & advisement of top NASA/JPL employees

**OHSU Neuroimage Lab** — **Graduate Research Assistant (Medical)** — **Oct 2017 – June 2018**

- Analyzed relationship of adjacent microbiome populations for menopause & reproductive health
- Developed processing/audit tools for survey & fMRI data to support ABCD study for ADHD/ASD
- Provided git, project management & security trainings for teams with varying readiness levels

**OHSU MSc Research** — **Machine Learning (Marine Acoustics)** — **Sept 2017 – Dec 2019**

- Developed track explorer for 10YR continuous audio track, supporting Aloha Cabled Observatory
- Developed anomaly detectors for noisy marine acoustics, based on variational auto-encoders
- Measured effects of spectral subtraction & DL models to reduce noise & improve audio quality

**Galois Inc.** — **R&D Software Engineer (Security)** — **April 2014 – Dec 2015**

- Developed processing pipelines on local clusters for evaluator workflows with multiple SBIR/STTRs
- Setup brittle PPAML languages on 30+ participants' devices for two annual DARPA summer schools
- Co-authored secure internet voting feasibility study for non-technical, policy-focused audience

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**Oregon & Health Science University**
**Computer Science & Machine Learning**
Masters of Science

**Western Washington University**
**Computer Science & Cryptography**
Bachelor of Science, Math Minor